

MICA — evidence figures

Where each number comes from: written rules, learning, or both

The system has two kinds of parts, and every figure below says which kind produced its numbers. **Hand-rule** parts are instructions a person wrote down — “compare the structure to these known shapes,” “only act when confidence passes this bar.” They do exactly what they were told, nothing more. **Observation** parts were not told what to do — they learned by watching recordings of a person building, the way one learns a coworker’s habits. **Both** means a hand-written frame carrying learned content inside it.

Figure / finding	written-rule part	learned-by-watching part	type
Fig. 1 five designs	the losing baseline itself; the belief-update rule	the winner’s evidence judgments	both
Fig. 2 real pretraining	data tidying only	teacher: the human’s own recorded next actions	obs.
Fig. 3 disputed labels	sorting sessions into groups	the trained parts; builder’s label as the answer key	both
Fig. 4 seeing = knowing	counting seen blocks	the shape representation	obs.
Fig. 5 live case study	the belief-update rule	evidence judgments	both
Fig. 6 move prediction	the definition of “right”	the predictor, end to end	obs.
structure checker	entirely written rules	—	hand
safety gate + consent	entirely written rules (by design)	—	hand

The split is itself a result. **Hand-rule parts are where trust lives:** on builds they were written for, they are perfect (the matcher scores 1.0 on the known-truth corpus) and anyone can read exactly why they answered as they did — but on new, unplanned builds they collapse (0.042 in Fig. 1), and that collapse was measured, not assumed. **Learned parts are where flexibility lives:** they keep working on builds nothing was written for (0.375), and they get better simply by watching more (Figs. 2–4). The system divides the work accordingly: *written rules decide when it is safe to speak or act; learned parts do the watching and predicting*. Fig. 1 is the head-to-head test behind that division — rules alone fail exactly where real, unscripted work happens.

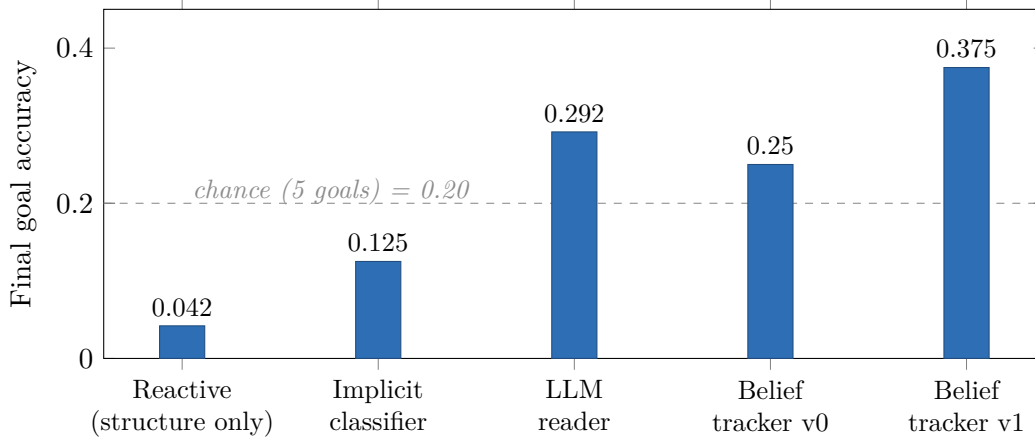


Figure 1: **Guessing what the builder is making: five designs, one test.** Five different designs watched the same 24 recorded building sessions — real, free-form builds that none of them had ever seen — and each had to name what the builder was making, out of five goal categories. Each bar is the share of sessions a design got right; random guessing scores 0.20 (dashed line). The leftmost design, which skips inference and just compares the structure to a library of known shapes, scores 0.042 — *worse* than guessing, because free-form builds match nothing in its library. The design that maintains a running belief about the builder’s intention scores 0.375, nine times better. Since every design saw identical input, the gap can only come from the inference itself. (Comparison run of 2026-07-19.) *Source: BOTH — the failing baseline is the hand-written rules; the winner is learned content inside a hand-written frame.*

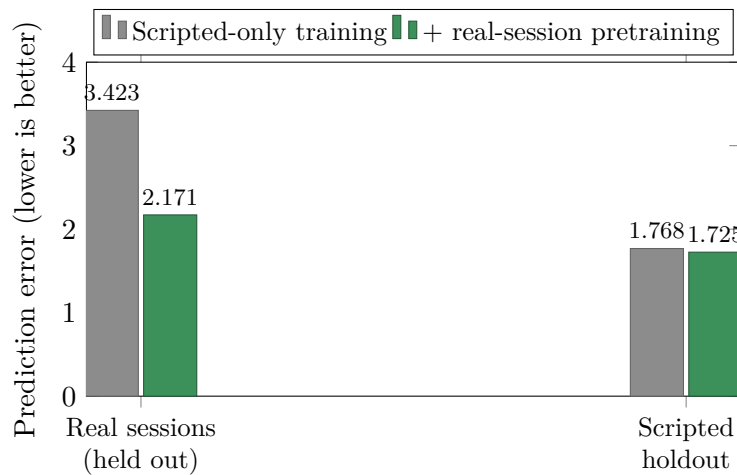


Figure 2: **A model trained on artificial data struggles with real people — until it watches real recordings.** The bars show prediction error when the model guesses a builder’s next action: lower is better. Trained only on computer-generated example builds (gray), the model predicts artificial builders well but real ones poorly — the left gray bar towers over the right one. Letting it first watch real recorded sessions (green) cuts its error on real builders by 37%. The key: this extra training needed *no labels at all*. The model simply watches a recording and tries to guess each next action, so even sessions whose goal was disputed or unidentifiable could teach it — a doubtful label cannot mislead training that uses no labels. The test sessions were locked in before the experiment ran. *Source: OBSERVATION — the teacher is the human’s own recorded next actions; written rules only tidy the data.*

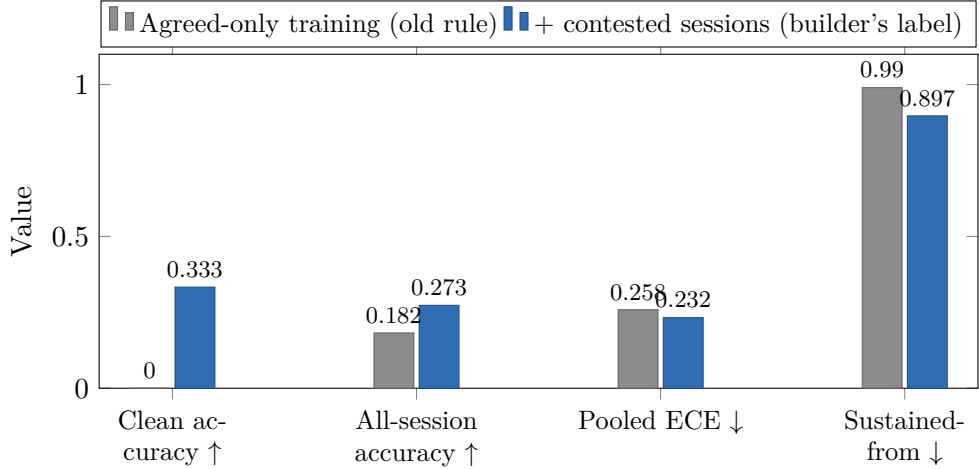


Figure 3: **When human and machine disagree about a session, trust the human.** Some recorded sessions are “disputed”: the builder said one goal, the automatic checker said another. The old rule kept such sessions out of training (gray bars); the new rule trains on them, taking the builder’s word as the truth (blue bars). On a shared test set that neither version trained on, keeping them improved all four measures: accuracy on cleanly-labeled sessions ($0 \rightarrow 0.33$), accuracy across all sessions, how well stated confidence matches actual correctness (ECE; lower is better), and how early in a build the right answer locks in and stays (lower is earlier). Honesty note: an earlier run of this same experiment found the *opposite*; a data flaw was found and repaired, the experiment was rerun, and the reversal is on the record — and re-checked at every retrain. The test set is small (11 sessions). *Source: BOTH — learned parts trained on watched sessions, with the builder’s own label as the answer key; written rules only sort sessions into groups.*

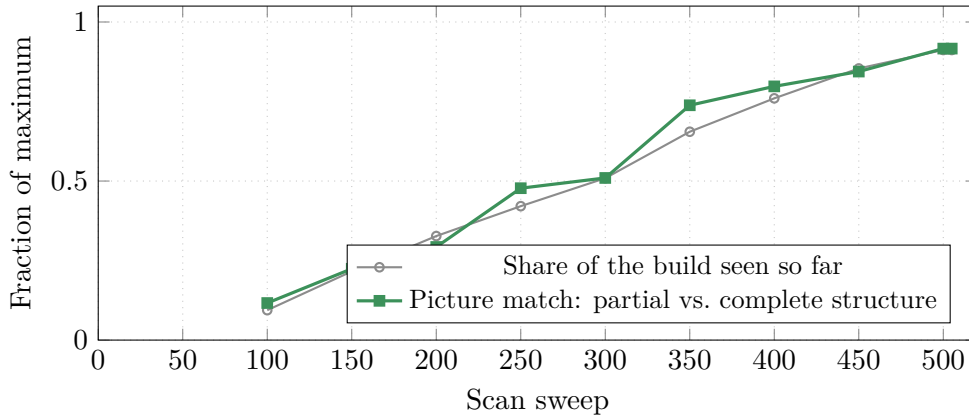


Figure 4: **Seeing more means understanding more.** The agent walks around the builder’s work and can only know what its own eyes have passed over — it gets no map, no list of blocks from the game. The gray line shows how much of the build it has seen so far; the green line asks a harder question: does its mental picture of the *partial* structure it has seen match the picture it would form of the *complete* structure? (1.0 = identical.) The two lines rise together: by the time 91% of the build has been seen, the match reaches 0.92 — watching most of the build is nearly as good as seeing all of it. (One fully instrumented session, `fabric-20260719-052914`.) *Source: OBSERVATION — the “mental picture” is a learned shape representation; only the seen-block counting is hand bookkeeping.*

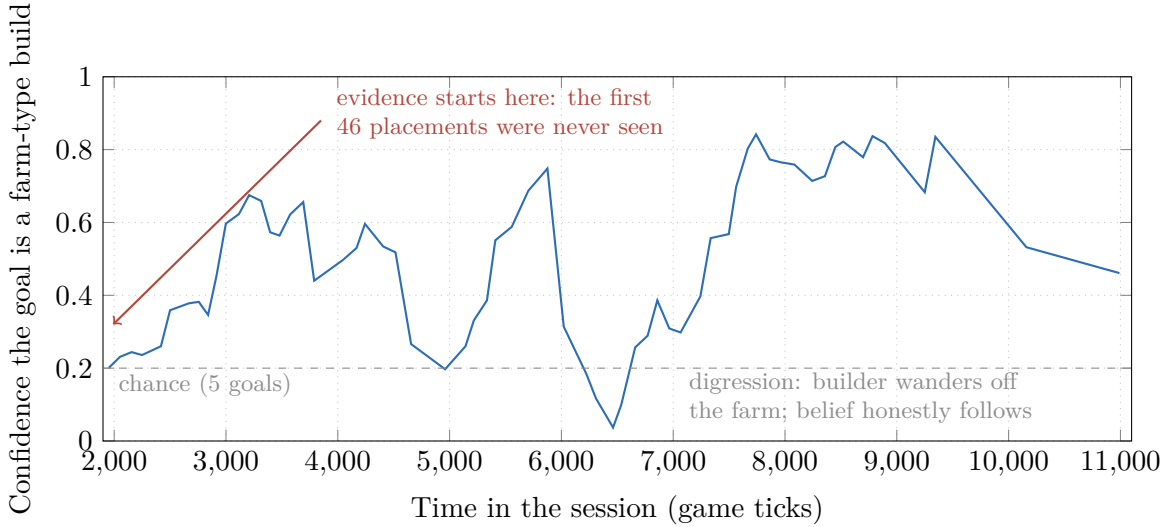


Figure 5: **One real session, moment by moment: the system changing its mind as evidence arrives.** The builder was making a crop farm (their own label, given after the session). The line is the system’s live confidence, at each moment, that “farm-type build” is the right answer. It starts at chance — the system connected late and missed the first 46 placed blocks, so at that point it genuinely knew nothing. As evidence arrives, confidence climbs. Mid-session it *drops*: the builder wandered away from the farm, and the system honestly followed the evidence away from the right answer rather than clinging to it. When building resumed, confidence re-locked and peaked at 0.84. One illustrative session — the statistical claims live in Fig. 1, not here. *Source: BOTH — a hand-written update rule deciding how beliefs change, fed by learned judgments of what the evidence suggests.*

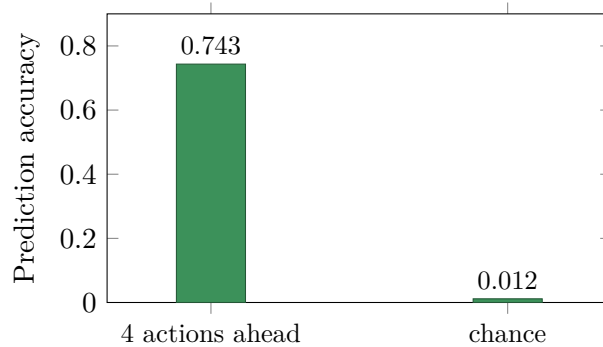


Figure 6: **Not just naming the goal — predicting the moves.** Every builder action is written in a fixed vocabulary of 87 possible moves (walk, turn, place this block, break that one, ...). Here the model must predict the builder’s action *four moves into the future*, on sessions kept out of training. It is right 74% of the time; blind guessing would be right 1.2% of the time (the small bar). “Right” means the exact move, with a small tolerance of ± 2 blocks on positions. Predicting several moves ahead is the practical test of understanding: it is what would let a helper prepare before being asked. *Source: OBSERVATION — the predictor is learned entirely from watching; only the definition of “right” is hand-written.*